How to find the standard error?

$$\bar{x}_1, \bar{x}_2, \bar{x}_3, \ldots, \bar{x}_k \qquad N \sim \left(\mu, \frac{\sigma^2}{n}\right)$$

Variance

$$\text{Standard deviation} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

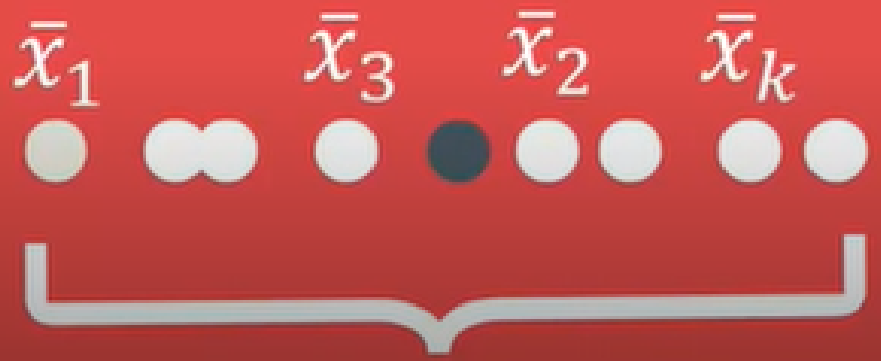


Meaning of the standard error



Like any standard deviation, it shows variability

Meaning of the standard error



Variability.. of the sample means

Meaning of the standard error

Used in most statistical tests



Because it shows how well you approximated the true mean

Note



$$\frac{\sigma}{\sqrt{n}}$$

n

Standard error decreases when sample size increases

Bigger sample - better approximation